SENTIMENT ANALYSIS OF DRUG REVIEWS

UCI ML Drug Review dataset

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Presentation Outline

Healthcare meets IT

- Major goals in the health and pharmaceutical industry is to ensure the effectivity and safety of drugs.
- Pharmaceutical product safety depends on clinical trials and specific test protocols.
- Such studies are typically done under strict conditions
  - Limited number of test subjects
  - Limited time span.
- User reviews as a resource offer great potential in obtaining such data.

Features in the dataset

DRUG
(CATEGORICAL)

CONDITION
(CATEGORICAL)

REVIEW
(TEXT)

USEFUL_COUNTS
(NUMERICAL)

RATING [1-10]
(NUMERICAL)

PREDICTOR

TARGET VARIABLE

Duplication alert!

- Scraping error in data collection caused the dataset to have a large amount of duplicates (not addressed in the research paper!).
- Duplicates = same text input repeated twice
  => biased model.
- After removal, dataset shrank almost by half.

Main predictive feature: patient reviews

POSITIVE REVIEW

"Absolutely saved my life. I've tried numerous drugs over 15 years and this is a miracle. No side effects whatsoever. I feel better than I ever have in my life".

NEGATIVE REVIEW

"Absolutely terrible experience on Tirosint - landed in ER with chest pains, dizziness, extremely disoriented."

Drug types and medical conditions in the dataset

3,199

UNIQUE DRUGS

837

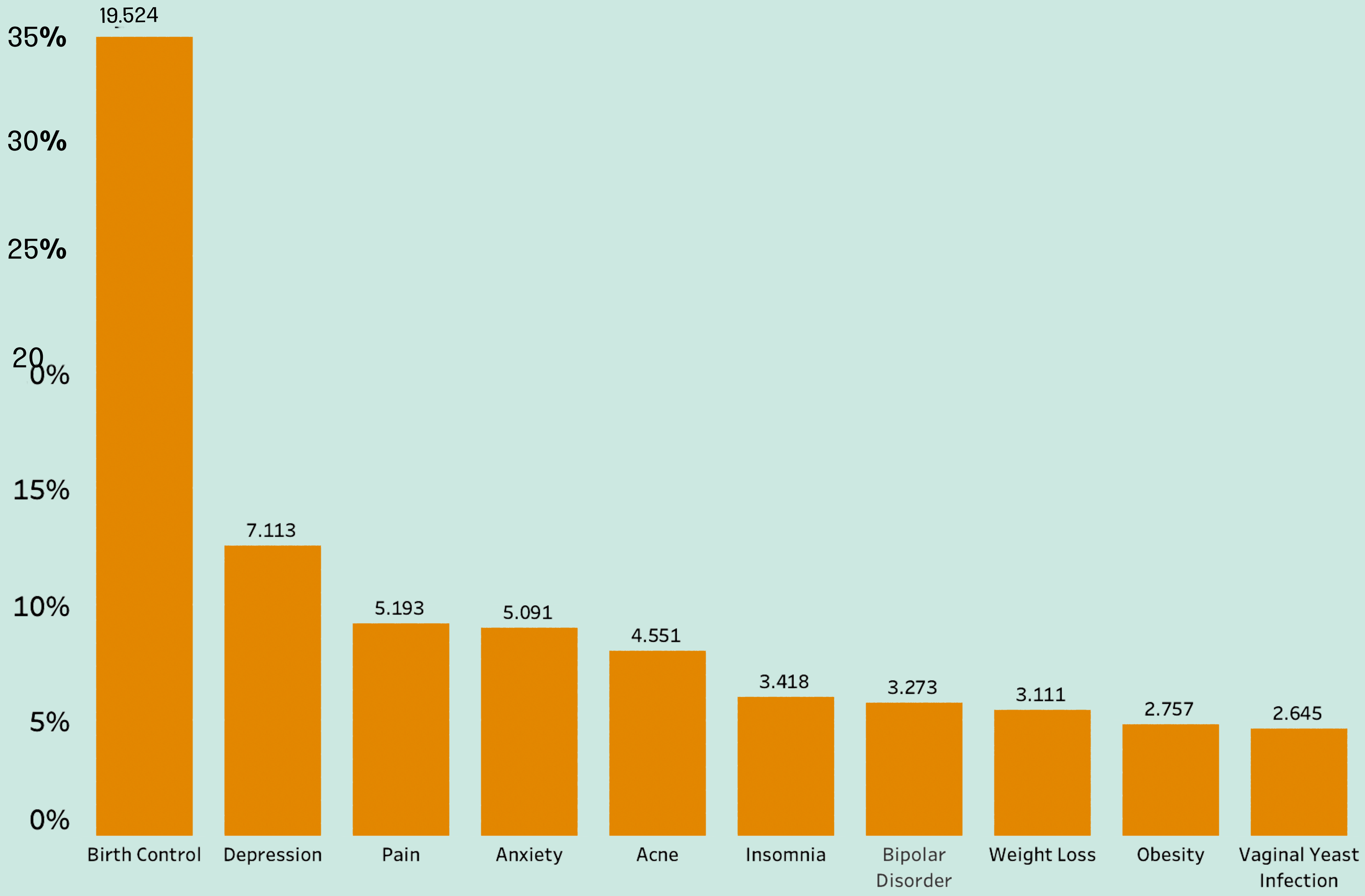
UNIQUE CONDITIONS

130,285

UNIQUE REVIEWS

Drug types and medical conditions in the dataset

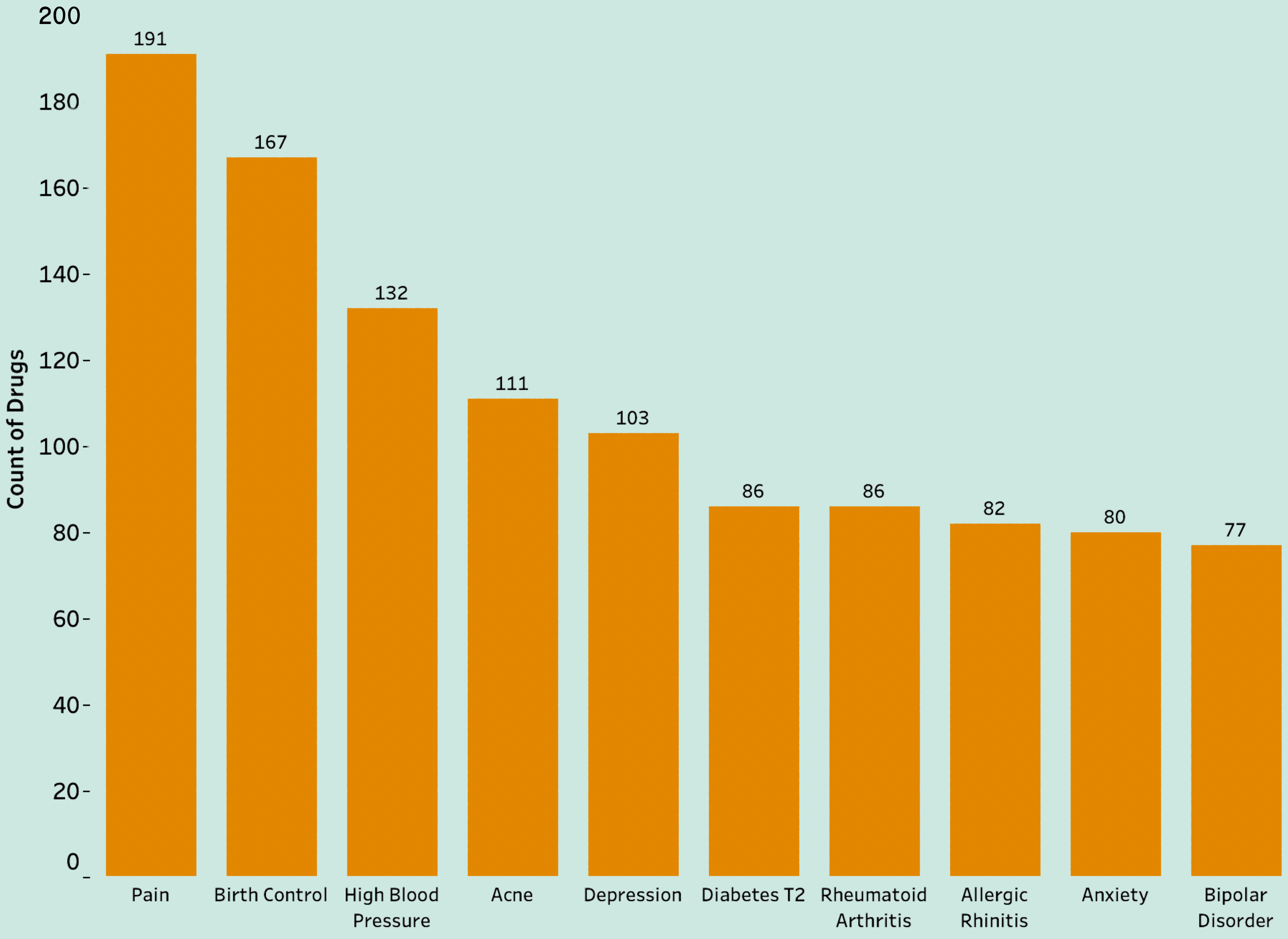
Top 10:
Review count per Condition



DRUG DATASET

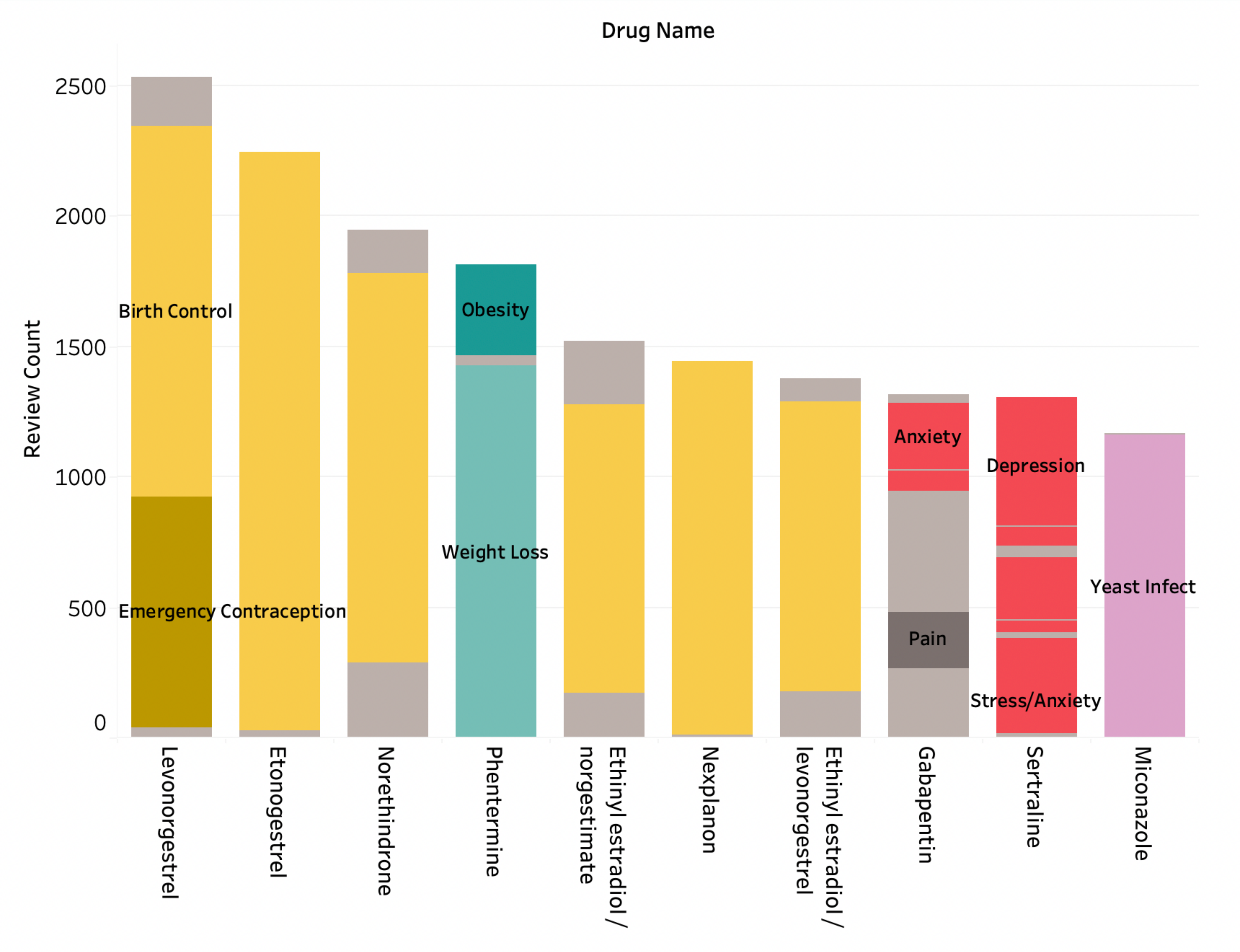
Drug types and medical conditions in the dataset

Top 10:
Drug count per Condition



Drug types and medical conditions in the dataset

Top 10 reviewed drugs, with most common target Condition

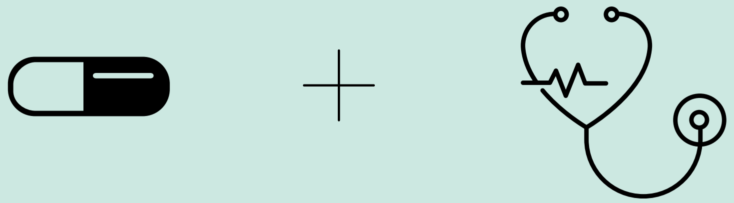


Most common conditions targeted by top 10 drugs in the dataset:

- Birth control and termination
- Mood disorders
- Weight loss

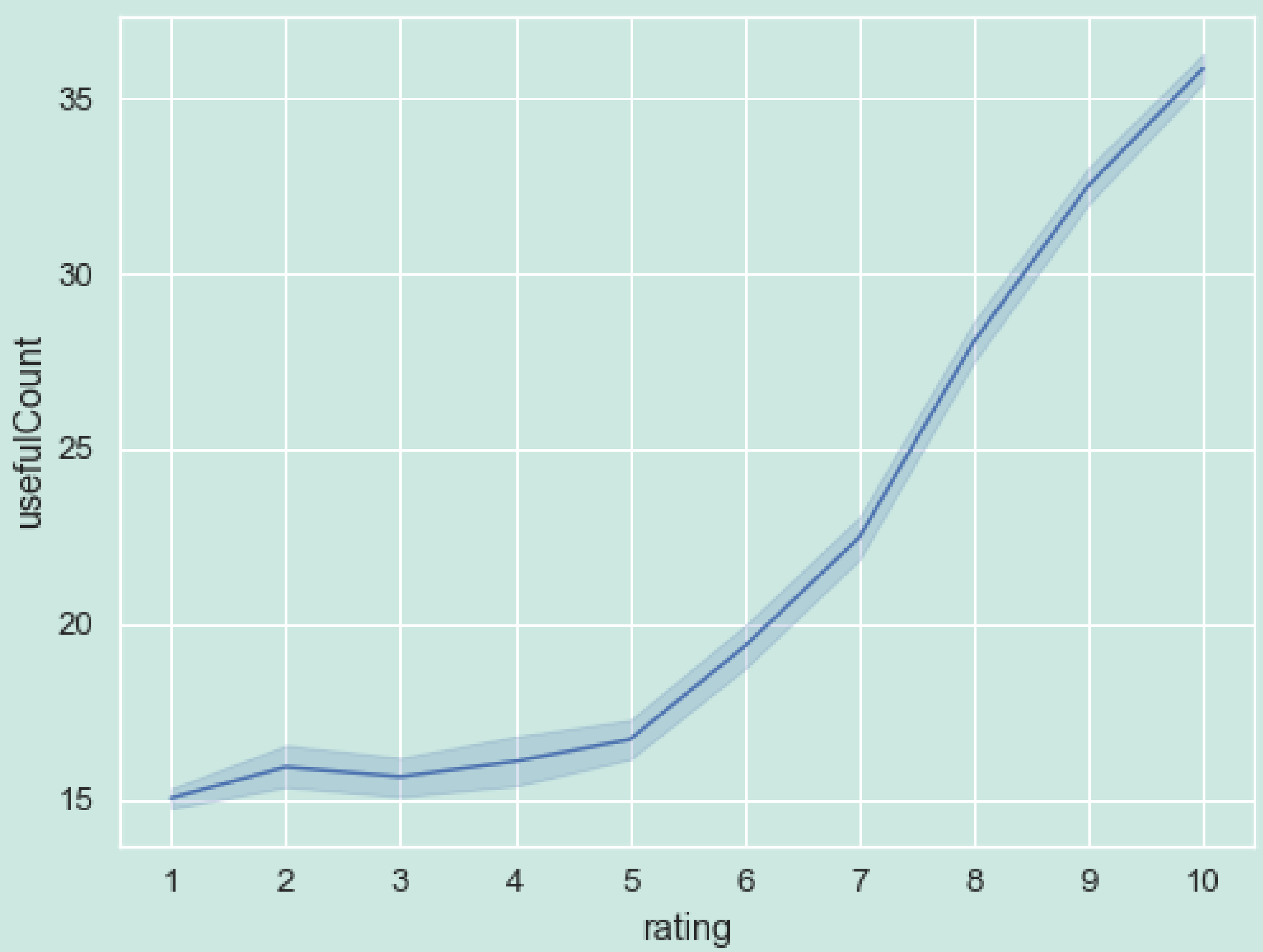
Drug recommender

The dataset allows users to find the best and worst drugs for the condition they are suffering from



Usefulness votes

Average of UsefulCount per Rating



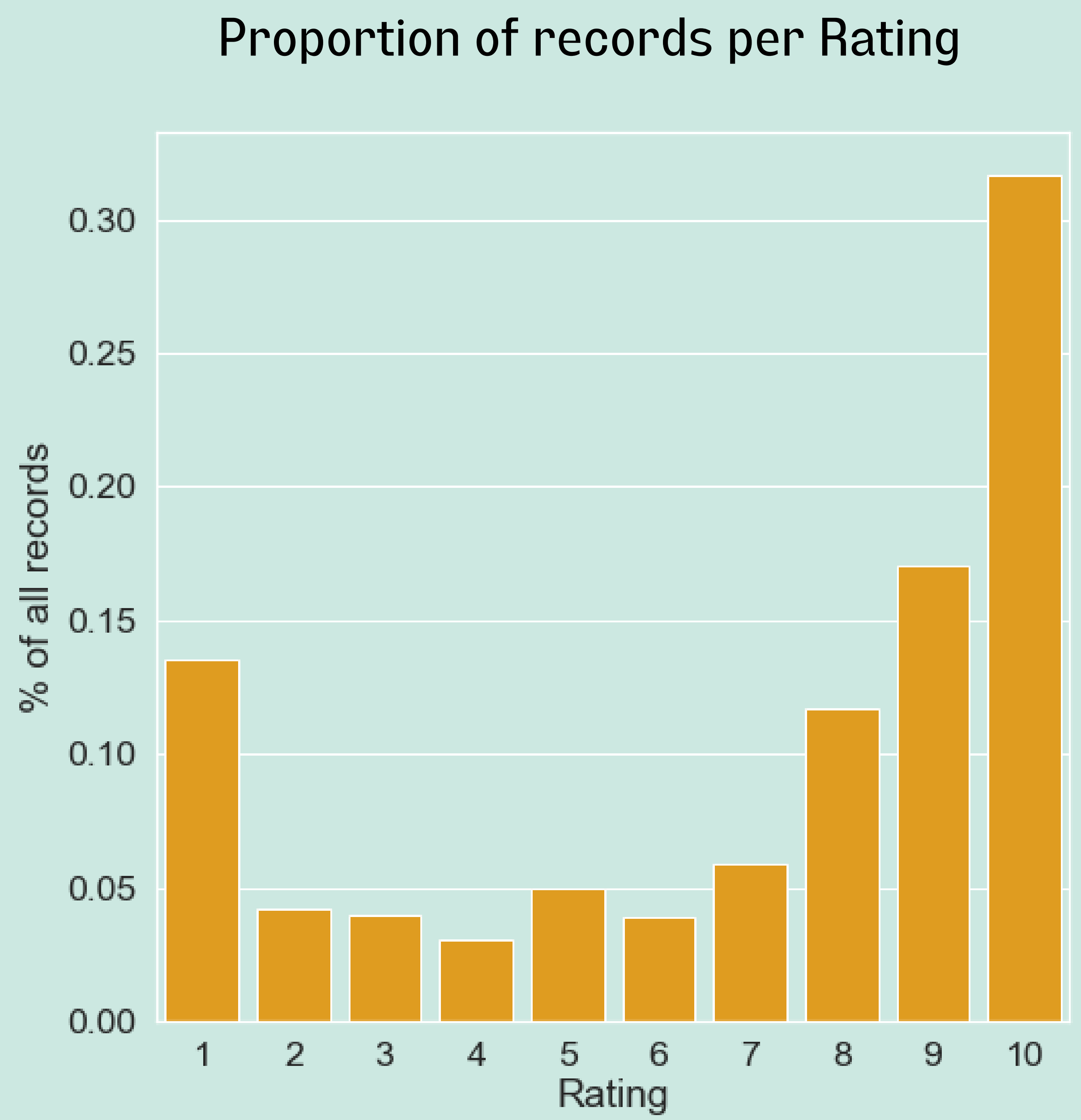
Comments associated with positive rating tend to receive more 'useful comment' votes.

However, data is imbalanced towards positive reviews (see next slide):

- Removing less useful ones would decrease negative reviews
- i.e. **increase imbalance**.

=> No removing of records

★★★★★ Target variable: drug rating



Feature engineering:

Rating of 1 - 6 ⟶ "**0**" class (negative/neutral)

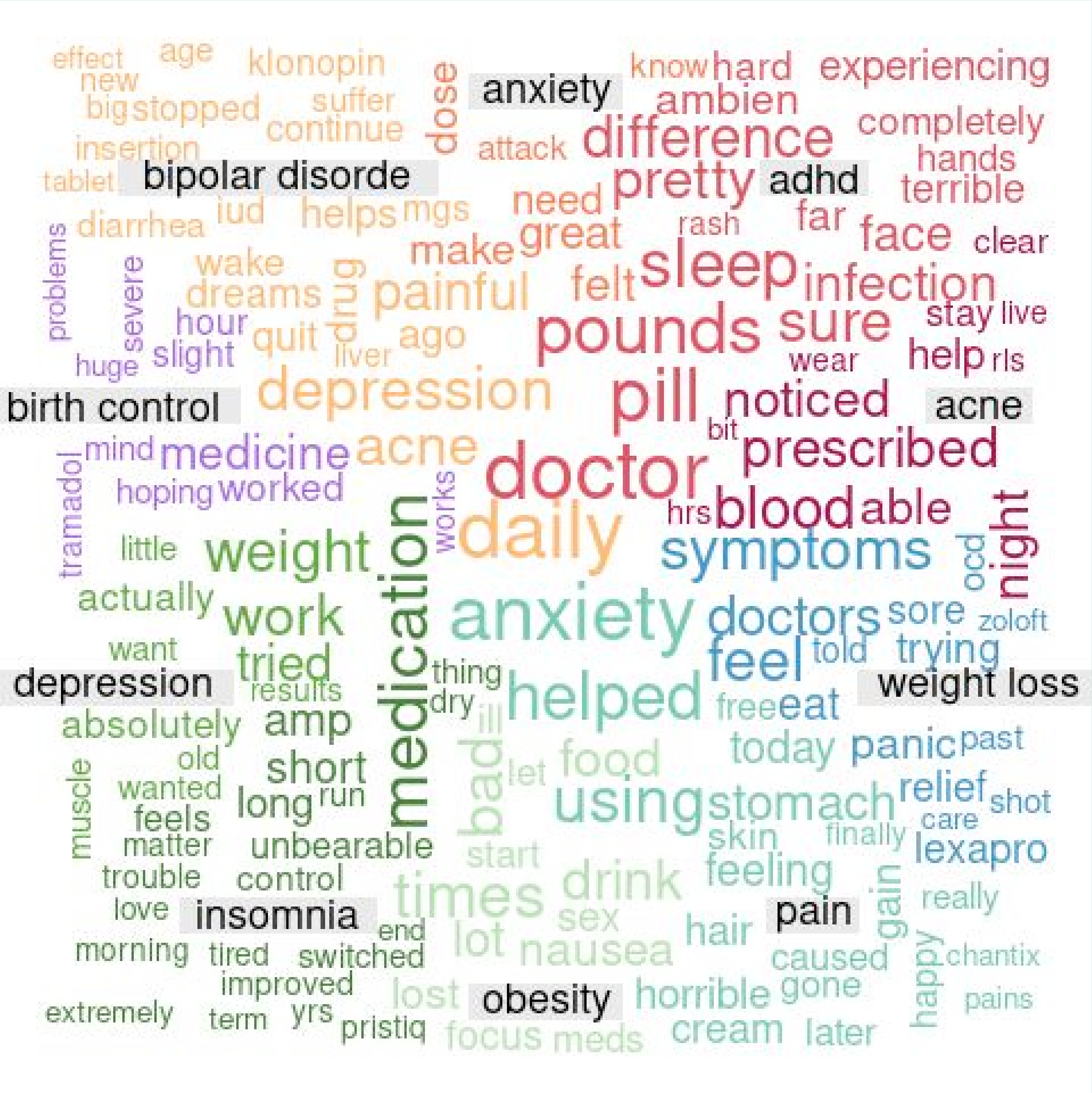
Rating of 7 - 10 ⟶ "**1**" class (positive)

Imbalanced data

Class 0 still twice as big, so I ran the models on both a full and a downsample.

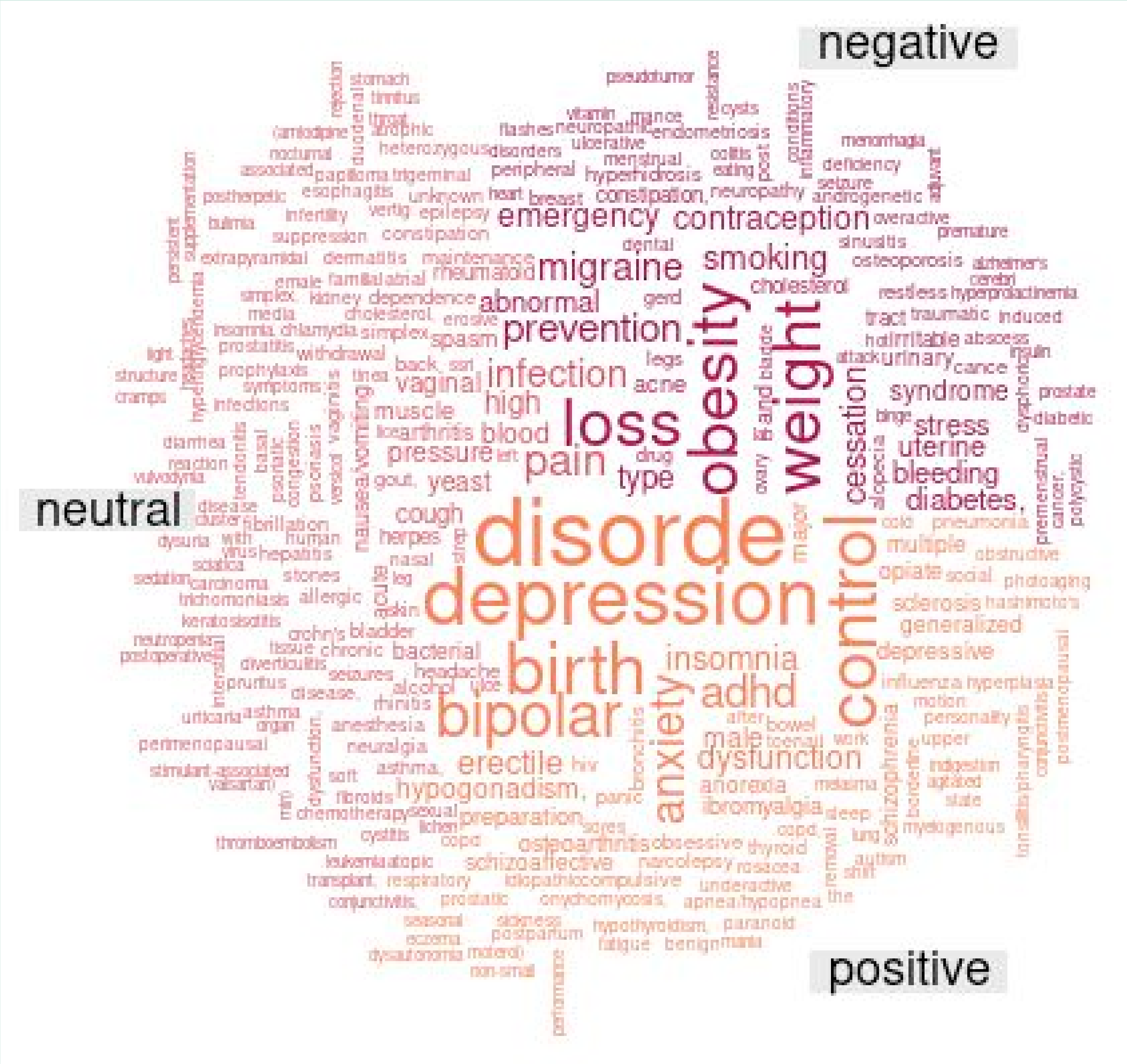
Word Clouds

Most frequent words associated with top 10 Conditions



Word Clouds

Reviews compared against word lists associated with positive / neutral / negative items



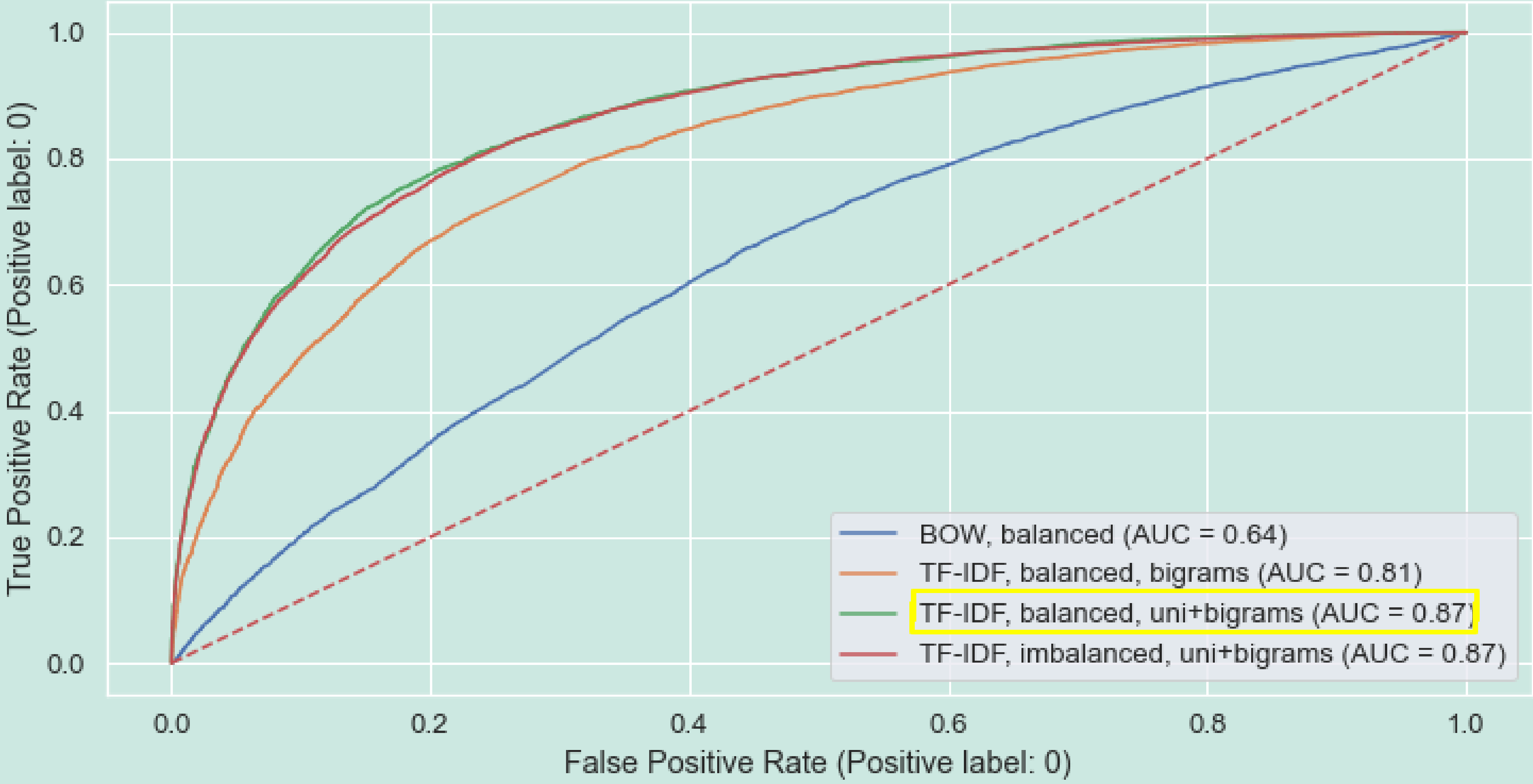
Model Comparison

- Using Random Forests

1. TF-IDF > BOW
2. Downsample > full sample
   - (Better recall for both categories)
3. Uni + bigrams > bigrams only

Overall Best model:

TF-IDF + uni & bigrams [balanced sample]



Balanced

Results obtained for the TRAIN SET
==================================
The Cohen's Kappa is: 0.96

	precision	recall	f1-score	support
0	0.98	0.98	0.98	34967
1	0.98	0.98	0.98	35194
accuracy			0.98	70161
macro avg	0.98	0.98	0.98	70161
weighted avg	0.98	0.98	0.98	70161

==================================
Results obtained for the TEST SET
The Cohen's Kappa is: 0.57

	precision	recall	f1-score	support
0	0.79	0.79	0.79	8884
1	0.78	0.78	0.78	8657
accuracy			0.79	17541
macro avg	0.79	0.79	0.79	17541
weighted avg	0.79	0.79	0.79	17541

Imbalanced, more 1s!

Results obtained for the TRAIN SET
==================================
The Cohen's Kappa is: 0.99

	precision	recall	f1-score	support
0	1.00	0.99	0.99	35026
1	0.99	1.00	1.00	69202
accuracy			1.00	104228
macro avg	1.00	0.99	0.99	104228
weighted avg	1.00	1.00	1.00	104228

==================================
Results obtained for the TEST SET
The Cohen's Kappa is: 0.46

	precision	recall	f1-score	support
0	0.84	0.45	0.58	8825
1	0.77	0.96	0.85	17232
accuracy			0.78	26057
macro avg	0.81	0.70	0.72	26057
weighted avg	0.80	0.78	0.76	26057

Conclusions and Future Ideas

- General:
  - Most common conditions people review on are **birth control**, **mood disorders**, **pain**, **obesity,** and **skin conditions**.
  - The current dataset allows us to identify medical conditions which are relatively well treated (e.g. depression), and target ones which are not (e.g. weight loss, female hair loss), and require more research.

- Sentiment Analysis:
  - Advantage: allows a much more fine grained assessment of drugs and medical conditions, discovering new associations previously unexpected.
  - Would improve with defining a 'neutral' category (did not make it to incorporate in actual model).
  - Use models which allow more semantic mapping of words than BOW/TF-IDF allow (Word2Vec).